# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** returns along the same path | For 90°, reversed. |
| 2 | **(b)** umbra | Complete shadow. |
| 3 | **(b)** virtual image position follows the law of reflection | Geometric consequence. |
| 4 | **(a)** increases by 15° | Twice the turn of the mirror (10 x 2). |
| 5 | **(b)** shadow | Dark. |
| 6 | **(c)** red | Same colour. |
| 7 | **(d)** 1.6 m tall | Half. |
| 8 | **(d)** double | Distance ratio 2:1. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | Because light cannot bend around the opaque object, a dark region (shadow) forms behind it. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are true; the Moon only reflects sunlight, but the luminous nature of the Sun does not explain it. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Angle of incidence is 0° (1); The ray retraces its path (1). | 1+1 |
| 12 | The Sun is low in the evening (1); Rays fall at a smaller angle so the shadow stretches (1). | 1+1 |
| 13 | It reflects sunlight (1); Surface scatters light (1). | 1+1 |
| 14 | 5 cm (1); Virtual, erect and laterally inverted (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1). | 1+1+1 |
| 16 | Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1). | 1+1+1 |
| 17 | Water gives regular reflection (1); Light from headlights bounces back (1); An image of the lights is seen (1). | 1+1+1 |
| | **OR:** Initially 8 m (1); Now 3 m, so 6 m (1); Decrease 2 m (1). | 1+1+1 |
| 18 | Reflect headlights (1); Improve visibility (1); Safety (1). | 1+1+1 |
| | **OR:** No light (1); Nothing to reflect (1); Need a source (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) A source of light, an opaque object and a screen (1). (ii) It becomes larger (1). (iii) Translucent (1). (iv) No (1), because clear glass is transparent and allows almost all the light to pass through (1). | 1+1+1+2 |
| 20 | Place object at several distances (1); Find image position by parallax with pin (1); Record distances (1); Compare in a table (1); Equal (1). | 1+1+1+1+1 |
| | **OR:** First mirror shows front (1); Second mirror behind (1); Reflect image of back (1); Repeated images (1); Customer sees the back (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 50 cm | 20 x 100/40. |
| 2 | **(c)** light from a lamp reflecting from the book | Absorbs red. |
| 3 | **(c)** 2 m behind the mirror | Equal distance. |
| 4 | **(a)** 50 m | 40/0.8. |
| 5 | **(b)** a still pond | Reflects red. |
| 6 | **(d)** 50° | Half. |
| 7 | **(d)** 9 m | 12 x 1.5/2. |
| 8 | **(a)** paper reflects light diffusely | Diffused reflection. |
| 9 | **(c)** A is true but R is false. | A is true; a virtual image cannot be obtained on a screen, so R is false. |
| 10 | **(d)** A is false but R is true. | The image is of the same size, so A is false; R is true. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Image of the Sun on a screen (1); Do not look at the Sun directly (1). | 1+1 |
| 12 | They give an erect image of the same size (1); Convex or concave would distort (1). | 1+1 |
| 13 | From 6 m to 10 m (1); Increases by 4 m (1). | 1+1 |
| 14 | The rays do not actually meet at the image (1); It cannot be taken on a screen (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Height ratio = 15/30 (1); 0.5 x 6 (1); 3 cm (1). | 1+1+1 |
| 16 | The polished surface is smooth (1); It reflects light regularly (1); A rusty surface scatters light (1). | 1+1+1 |
| 17 | Write a word (1); Hold before a mirror (1); Letters reversed (1). | 1+1+1 |
| | **OR:** Ratio 1.2/0.9 (1); 30 x 1.333 (1); 40 m (1). | 1+1+1 |
| 18 | Place a screen behind the mirror (1); No image forms on it (1); Image seen only in the mirror (1). | 1+1+1 |
| | **OR:** Most light passes (1); A little is reflected and absorbed (1); Faint shadow (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) 2 m behind the mirror (1). (ii) $2 + 2 = 4$ m (1). (iii) It is virtual and erect, and of the same size as the object (1). (iv) Drivers see the front of the ambulance in their rear-view mirrors (1); lateral inversion makes the word look correct in the mirror so they can read it and give way (1). | 1+1+1+2 |
| 20 | Ray enters top mirror (1); Reflects at 45° (1); Goes down to second mirror (1); Reflects to eye (1); Longer tube sees further (1). | 1+1+1+1+1 |
| | **OR:** Magnification = 90/30 = 3 (1); Diameter 15 cm (1); Area ratio 9:1 (1); Rays straight (1); Check (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** diffused reflection | Scatters. |
| 2 | **(b)** very large | Repeated reflection. |
| 3 | **(d)** 4 cm | 200 x 20/1000. |
| 4 | **(d)** regular reflection | Water acts as a mirror. |
| 5 | **(c)** 1 cm | 20 x 10/200. |
| 6 | **(d)** double | Inversely. |
| 7 | **(d)** a shadow is darker where no light reaches | Shadow is absence of light. |
| 8 | **(d)** laterally inverted | Reverse. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | Crossing of the straight rays at the hole inverts the image. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are laws of reflection but one does not explain the other. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Rear-view mirror shows it correct (1); Easy to read (1). | 1+1 |
| 12 | A small source gives a sharp shadow (1); A large source gives an umbra and a penumbra (1). | 1+1 |
| 13 | Glare dazzles drivers (1); Reduces visibility (1). | 1+1 |
| 14 | Change angle between mirrors (1); Count the images (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ratio 12/6 = 2 (1); Distance = 2 x 20 (1); 40 cm (1). | 1+1+1 |
| 16 | Cut windows (1); Fix mirrors at 45° (1); Look in the lower window (1). | 1+1+1 |
| 17 | Place a pin in front (1); Look from different sides (1); Line up the second pin (1). | 1+1+1 |
| | **OR:** Image is of the Sun, not the hole (1); Small hole (1); Circular Sun (1). | 1+1+1 |
| 18 | i changes by 20° (1); r also changes by 20° (1); Angle between rays changes by 40° (1). | 1+1+1 |
| | **OR:** Image larger (1); Dimmer (1); Same inversion (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Inverted (1). (ii) Real, as it is formed on a screen (1). (iii) The image becomes brighter but blurred (1). (iv) Light travels in straight lines (1), so rays from the top of the flame go to the bottom of the screen and rays from the bottom go to the top after crossing at the hole (1). | 1+1+1+2 |
| 20 | The Sun is low in the sky (1); Rays fall at a small angle (1); The shadow stretches (1); At noon the Sun is high (1); The shadow is shortest (1). | 1+1+1+1+1 |
| | **OR:** Mirrors (1); Shiny utensils (1); Windows (1); Shadow on the wall (1); Conclusions (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
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